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Identification and characterization of novel ferroptosis-related genes in acute myocardial infarction

Qiaoyu Zhou¹, Ruizheng Shi², Jia Liu³ and Zhaoya Liu^{4*}

Abstract

Background Acute myocardial infarction (AMI) is a leading cause of death and morbidity worldwide. Ferroptosis, a form of regulated cell death, plays a critical role in modulating immune functions during AMI. This study aimed to identify ferroptosis-related hub genes that could serve as potential therapeutic targets in the progression of AMI.

Methods Bioinformatics was used to identify overlapping genes associated with ferroptosis and the infiltration of 22 immune cells by Cell-type Identification by Estimating Relative Subsets of RNA Transcript (CIBERSORT) analysis. The expression of ferroptosis-related genes in AMI was validated across independent datasets, clinical samples, and *in vitro* cellular experiments. The predictive value for heart failure was evaluated in the first dimension of principal component analysis (PCA) using receiver operating characteristic (ROC) analysis.

Results The study identified 11 key ferroptosis-related genes significantly correlated with immune cell abundance. CIBERSORT analysis highlighted immune dysregulation in AMI. *JDP2*, *DUSP1*, *TLR4*, *NFS1*, and *SLC1A5* were identified as potential biomarkers for AMI progression. Additionally, *JDP2*, *DUSP1*, and *DDIT4* demonstrated strong predictive value for long-term heart failure.

Conclusion This study highlights the potential association of ferroptosis-related genes with the pathogenesis of AMI, suggesting a role in the molecular mechanisms that may underlie acute coronary events.

Keywords Acute myocardial infarction, Ferroptosis, CIBERSORT, Immune function, Diagnostic biomarker genes, Bioinformatics

Introduction

Nearly 7.29 million people globally suffered from acute myocardial infarction (AMI) in 2015, which represented a marked aggravation of the global health situation [1, 2]. Despite advances in cardiovascular medicine, the molecular mechanisms of AMI remain unclear. With an aging global population, understanding these mechanisms is essential for discovering novel biomarkers and therapies to improve diagnosis, treatment, and survival in patients with acute coronary syndrome.

Ferroptosis, characterized by iron dependence and lipid peroxide buildup, is gaining attention as a key factor in AMI progression due to its role in membrane damage and redox imbalance [3–5]. Beyond the

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cellular implications, ferroptosis interferes with anti-tumor immunity and elicits immunogenic cell death [6]. Given the substantial evidence linking immune cell dysregulation with AMI progression [7–9], and the critical role of ferroptosis in ischemic injuries, particularly AMI, as indicated by its effects on infarct size and ventricular function [10–12], investigating ferroptosis as a potential therapeutic target is of paramount importance.

The advent and accessibility of advanced bioinformatics tools have revolutionized the identification of molecular markers and signaling pathways associated with specific diseases [13, 14]. The Cell-type Identification by Estimating Relative Subsets of RNA Transcript (CIBERSORT) is one such method that enables the quantification of cell types from bulk tissue gene expression profiles via RNA sequencing. It has been proven effective in accurately estimating immune cell profiles, as demonstrated in numerous studies [15–17]. In this study, we applied CIBERSORT to explore the immune landscape influenced by ferroptosis-related genes in the context of AMI.

Using the GSE60993, GSE29532 datasets, containing gene expression profiles of healthy individuals and AMI, we aim to identify potential ferroptosis-induced biomarkers and explicate the immune-modulatory role of these genes, thus contributing to a better understanding of the interplay between ferroptosis and immune response in AMI. Stable coronary artery disease (SCAD) is often regarded as a precursor to acute coronary events, such as AMI. While both conditions share similar underlying pathophysiological mechanisms (namely atherosclerosis), they exhibit distinct characteristics during the acute phase [18]. The European Society of Cardiology recently proposed renaming stable CAD as “chronic coronary syndrome.” However, this designation emphasizes similarities rather than differences between the acute and chronic presentations of CAD [19]. Management of SCAD centers on medication to prevent myocardial infarction and mortality [18]. Therefore, including SCAD as a control group allows for a more precise identification of biomarkers and molecular pathways specifically associated with the acute transition from stable disease to AMI. The dataset GSE59867, including individuals diagnosed with SCAD or AMI, and heart failure events, was employed for validating these findings. Besides, we conduct further experiments using cellular models and clinical samples, which will allow us to confirm the relevance of these biomarkers in a physiological context. This endeavor seeks to elucidate the influence of ferroptosis on the trajectory of AMI, offering avenues for optimized patient management and therapeutic strategies.

Materials and methods

Public datasets

From the Gene Expression Omnibus (GEO) database (<http://www.ncbi.nlm.nih.gov/geo/>), we downloaded RNA-seq data from the GSE60993, GSE29532, and GSE59867 datasets. The GSE60993 and GSE29532 datasets contained peripheral blood samples from 15 AMI patients and 14 healthy participants. The GSE59867 dataset contained information from the peripheral blood samples of patients with AMI ($n=111$) and control ($n=46$) groups comprising patients with SCAD. 259 ferroptosis-related genes were identified from the FerrDb database (<http://www.zhounan.org/ferrdb/current/>). In this study, GSE60993 and GSE29532 datasets were utilized for primary analyses in sections “Differential gene expression analysis” to “Analyzing signaling pathways and biological functions”, including differential gene expression, immune cell profiling, and pathway enrichment analysis. Subsequently, the independent dataset GSE59867 was employed in section “Single sample Gene Set Enrichment Analysis” to validate the key findings and predictive value for long-term heart failure to ensure robustness and generalizability.

Differential gene expression analysis

The “limma” package in R was used to determine differentially expressed genes (DEGs) by normalizing data, eliminating genes with high or low expression, and employing various statistical techniques including exact tests, empirical Bayes estimations, linear generalization models, and quasi-likelihood tests. The DEGs were selected using $P < 0.05$ and $|\log FC| > 0.5$ as the selection criteria.

Profiling of immune cells

To calculate immune cell proportions from bulk RNA-seq data using CIBERSORT, the data were formatted appropriately and uploaded onto the CIBERSORT website (<https://cibersort.stanford.edu/>) [17]. The p -values calculated by CIBERSORT for deconvolutions of each sample provided a measure of confidence. p -values of < 0.05 indicated statistical significance.

Principal component analysis (PCA)

To obtain attributes representative of the overall characteristics of complex matrices, PCA is commonly performed. We used PCA to assess whether a gene list could distinguish between the two groups in this study. Based on receiver operating characteristics (ROC) curves, the first dimension (Dim 1) was used to calculate the area under the curve (AUC) for a given gene list. AUCs of > 0.9 generally represent outstanding discrimination, while 0.8–0.9, 0.7–0.8, 0.5–0.7, and 0.5 indicate excellent,

acceptable, moderate, and no discrimination, respectively [20].

Analyzing signaling pathways and biological functions

The "clusterProfiler" package in R was used to analyze biological function and signaling pathways [21]. Gene ontology (GO) was used for functional annotation, and the Kyoto Encyclopedia of Genes and Genomes (KEGG) was used to identify enriched pathways [22–24]. Some signaling pathways could be missed in GO and KEGG analyses, and we additionally conducted a gene set enrichment analysis (GSEA) using "clusterProfiler". Pathway enrichment analysis was refined to focus on processes directly relevant to myocardial infarction, including oxidative stress response, inflammatory pathways, and cellular apoptosis mechanisms. Pathways unrelated to cardiac events, such as graft rejection, were excluded. The values of *p*AdjustMethod, *q*valueCutoff, *p*valueCutoff, *n*Perm, *max*GSSize, and *min*GSSize, which were 'BH', 0.2, 0.05, 1000, 500, and 10, respectively, were considered statistically significant.

Single sample gene set enrichment analysis

A total of 3049 human signature gene sets were downloaded from the Molecular Signature Database (<https://www.gsea-msigdb.org/gsea/msigdb/index.jsp>) on September 20, 2022. First, pathways that were significantly enriched in the GSEA analysis (*p*-value < 0.05, false discovery rate (FDR) < 0.25) were considered. Second, a literature review was performed to further refine this list, ensuring that only those pathways directly linked to immune function and inflammation in the context of AMI were included. Among them, 25 immune-related signaling pathways were selected for subsequent analysis. Single sample Gene Set Enrichment Analysis (ssGSEA) was employed to quantify the expression levels of 25 selected immune-related signaling pathways in the AMI and the control groups. ssGSEA is a variation of the GSEA algorithm that instead of calculating enrichment scores for groups of samples (i.e. Control vs Disease) and sets of genes (i.e. pathways), it provides a score for each sample and gene set pair.

Identification of the ferroptosis-related genes for AMI using an independent dataset

Further analysis and validation in this section were performed using an independent dataset, GSE59867 (<https://www.ncbi.nlm.nih.gov/geo/>) (Supplementary Table S1). The dataset included 94 patients with AMI at admission and 46 patients with SCAD, and the gene expression levels for 11 candidates from the two groups of patients were compared. We also calculated the AUCs for the ROC curves. The ferroptosis-related hub genes

with AUCs of ≥ 0.70 ($P < 0.05$) were considered to have a high predictive value and effectively distinguished AMI from long-term heart failure with remarkable specificity and sensitivity.

Cell line and treatment

The AC16 Human Cardiomyocyte Cell Line (ATCC; Rockville, MD, USA) is a widely used *in vitro* model for simulating ischemic heart disease in numerous studies [25–27]. AC16 cells were cultured in DMEM/F12 medium containing 10% fetal bovine serum at 37°C in an incubator with 5% CO₂ as the normoxic condition. To imitate the myocardial ischemia, AC16 cells were incubated in a hypoxic condition with 94% N₂, 5% CO₂, and 1% O₂ for 24 h to stimulate hypoxia.

Real-time reverse transcription PCR verification of clinically relevant genes

Following the Declaration of Helsinki, all protocols and procedures for the collection of human blood samples used in this study were approved by the Institutional Review Board of the Xiangya Hospital of the Central South University. The presence of coronary artery occlusion or significant stenosis was proved by emergency coronary angiography. Five patients with AMI were compared with five healthy individuals. Supplementary Table S2 provides the clinical characteristics of the patients. Peripheral blood mononuclear cells (PBMCs) were obtained from patients with AMI and healthy controls using density gradient centrifugation with Ficoll reagents from Cytiva (America). As directed by the manufacturer, RNA extraction, reverse transcription, and real-time PCR were carried out using the Evo M-MLV RT Mix Kit, SYBR® Green Premix kit, and RNAex Pro reagent (Accurate, Changsha, China), respectively. Supplementary Table S3 lists the primer sequences.

Statistical analyses

The R (version 4.0.1) language was used for all statistical analyses. Shapiro–Wilk and Kolmogorov–Smirnov tests were used to test the normality of the distributions. For normally distributed data, *t*-test was used, and for all other data, the Mann–Whitney U test was used. The correlation between two continuous variables was determined using Pearson's correlation coefficients. Statistical significance was denoted by $P < 0.05$.

Results

Data preprocessing and quality control

To achieve a robust and representative sample size of patients with acute myocardial infarction (AMI), we merged two datasets from the Gene Expression Omnibus (GEO), namely GSE60993 and GSE29532, as detailed

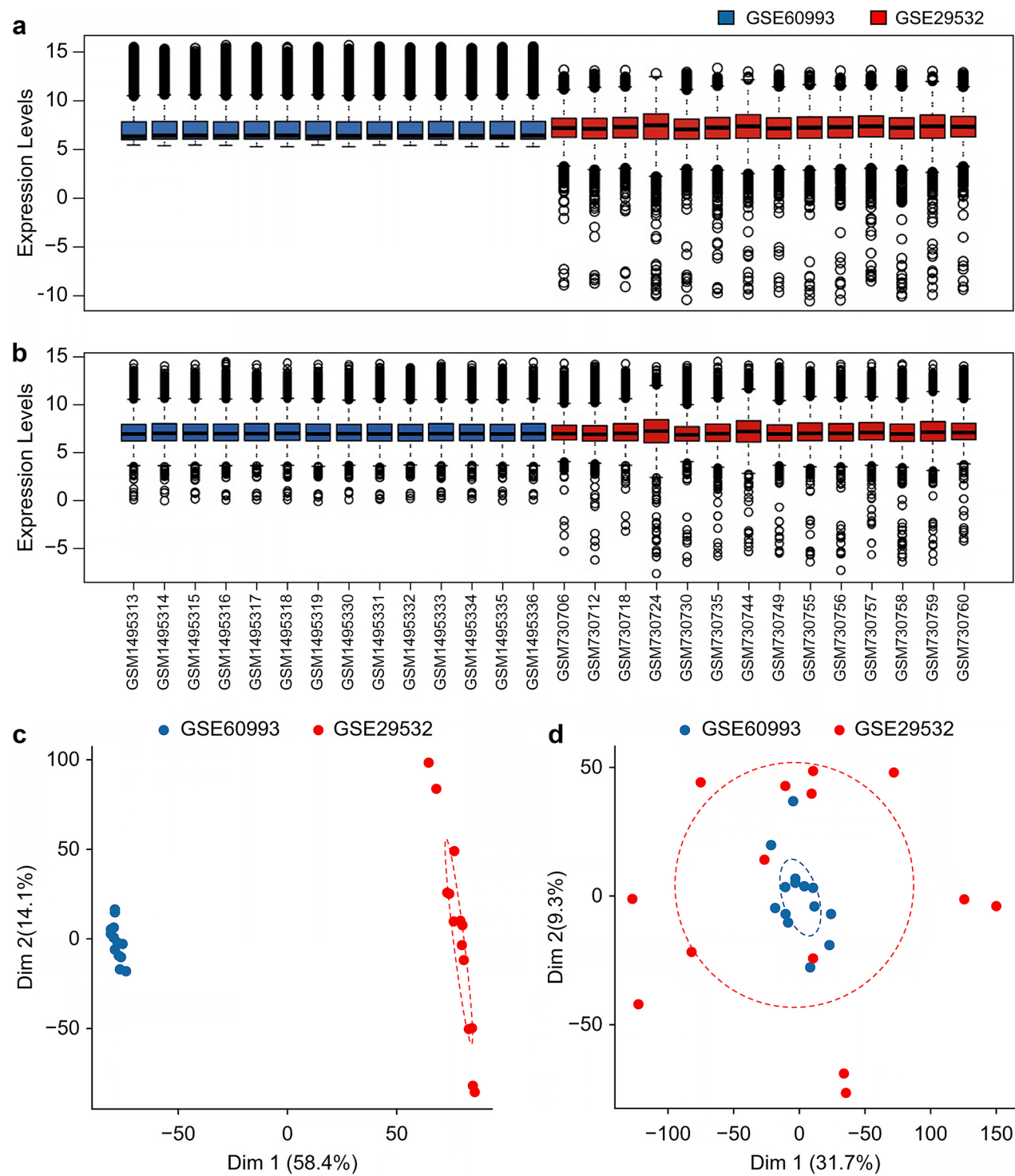


Fig. 1 Preprocessing and quality control of the data. **(A)** Batch effects were found in the two datasets. **(B)** The batch effects were successfully removed. **(C)** Before the removal of the batch effects, the two cohorts were separated. **(D)** After the removal of batch effects, the two datasets were merged. *Data source: merged data of GSE60993 and GSE29532 datasets from the GEO database

in Supplementary Table S4. We assumed that the two cohorts had batch effects, and we evaluated the levels of expression (Fig. 1A). Utilizing a linear model approach, we successfully eliminated these batch effects, achieving a normalized and integrated dataset, as illustrated in (Fig. 1B). The presence of batch effects in AMI patients was further assessed using PCA. Prior to the removal of batch effects, the two cohorts displayed distinct separation (Fig. 1C). Following batch effect correction, the datasets were effectively integrated, confirming successful normalization (Fig. 1D).

Identification of key ferroptosis-related genes

To identify critical ferroptosis-related genes, we performed a differential expression analysis comparing patients with AMI against healthy controls (Fig. 2). We identified 446 differentially expressed genes (DEGs) 201 upregulated, 245 downregulated with $P < 0.05$, $|\log FC| > 0.5$ (Supplementary Table S5, Fig. 2A). PCA further highlighted distinct expression patterns between AMI patients and controls, with notable separation along Dim 1 (Fig. 2B). A heat map provided a visual representation of these differential gene expressions (Fig. 2C).

To narrow our focus to ferroptosis-related genes, we performed an intersection analysis between the 446 DEGs and 259 ferroptosis-related genes, identifying 11 key genes: dual-specificity protein phosphatase 1 (*DUSP1*), toll-like receptor 4 (*TLR4*), ferritin heavy chain 1 (*FTH1*), solute carrier family 40 member 1 (*SLC40A1*), DNA damage-inducible transcript 4 (*DDIT4*), Jun dimerization protein 2 (*JDP2*), lysosome-associated membrane glycoprotein 2 (*LAMP2*), cysteine desulfurase (*NFS1*), neutral amino acid transporter 5 (*SLC1A5*), peroxiredoxin-1 (*PRDX1*), and X-box binding protein 1 (*XBPI*). These genes displayed significantly different expression levels between AMI patients and controls (Fig. 2D).

Potential key signaling pathways associated with AMI

The distinct gene expression patterns observed in AMI patients and controls prompted further investigation into the signaling pathways involved in AMI pathogenesis. We conducted GO and KEGG pathway enrichment analyses on the 201 upregulated and 245 downregulated genes. The top ten enriched biological processes included T-cell activation and leukocyte-cell adhesion (Fig. 3A). Notably, cell adhesion and differentiation of Th1, Th2, and Th17 cells emerged as consistently upregulated pathways (Fig. 3B). GSEA indicated enrichment of pathways such as aminoacyl-tRNA biosynthesis, the citrate cycle, and DNA replication, suggesting their active role in the cellular processes associated with AMI, including potential immune responses (Fig. 3C and D, Supplementary Table S6).

Profiling of the immune microenvironment in AMI

Given the enrichment of immune-related signaling pathways, we proceeded to analyze the immune cell abundance in AMI patients versus controls. The heat map illustrated differential patterns of immune cell expression (Fig. 4A). Significant differences were noted in the infiltration levels of B cells, CD8 T-cells, macrophages, and neutrophils between the two groups ($P < 0.05$), as detailed in Supplementary Table S7 and shown in (Fig. 4B). Correlation analysis revealed that specific ferroptosis-related genes, such as *LAMP2*, *JDP2*, *SLC40A1*, *FTH1*, *TLR4*, and *DUSP1*, were positively correlated with macrophage abundance, while genes like *XBPI*, *PRDX1*, *SLC1A5*, and *NFS1* showed negative correlations (Fig. 4C).

Profiling of immune-related signaling pathways in AMI

To further investigate changes in immune functions in AMI, we analyzed the relative expression levels of 25 immune-related signaling pathways selected from 3049 canonical pathways (Supplementary Table S8). Among these, eight pathways were significantly downregulated in AMI, including pathways involved in immunoregulatory interactions between lymphocytes and non-lymphocytes, primary immunodeficiency, and the adaptive immune system (Fig. 5A). Boxplot analysis showed distinct differences in these pathways between controls and AMI patients (Fig. 5B). Furthermore, the 11 identified ferroptosis-related genes were significantly associated with changes in these immune-related pathways (Fig. 5C).

Validation and predictive value of ferroptosis-related genes in AMI and heart failure

The identified ferroptosis-related genes, which showed differential expression between controls and AMI patients, were also strongly linked to immune function. Validation of these findings was performed using the GSE59867 dataset, and ROC curve analysis was conducted (Supplementary Table S9). Figures (6A–E) demonstrate significant differential expression of genes such as *DUSP1*, *JDP2*, *TLR4*, *NFS1*, and *SLC1A5* between patients with SCAD and AMI, with p -values of 0.0011, 0.0003, < 0.0001 , < 0.0001 , and 0.0002, respectively. These genes were consistently upregulated in AMI cohort. Furthermore, the ROC curves confirmed their diagnostic potential, with AUC values of 0.6610, 0.6759, 0.7821, 0.7599, and 0.6762, respectively (Fig. 6F). Based on the bioinformatics analysis of the GSE59867 dataset, we prioritized these five genes for further experimental validation due to their significant differential expression and potential clinical relevance. The high levels of upregulation and their promising diagnostic capabilities support their selection, while genes that did not meet these criteria were not pursued for validation. To substantiate our

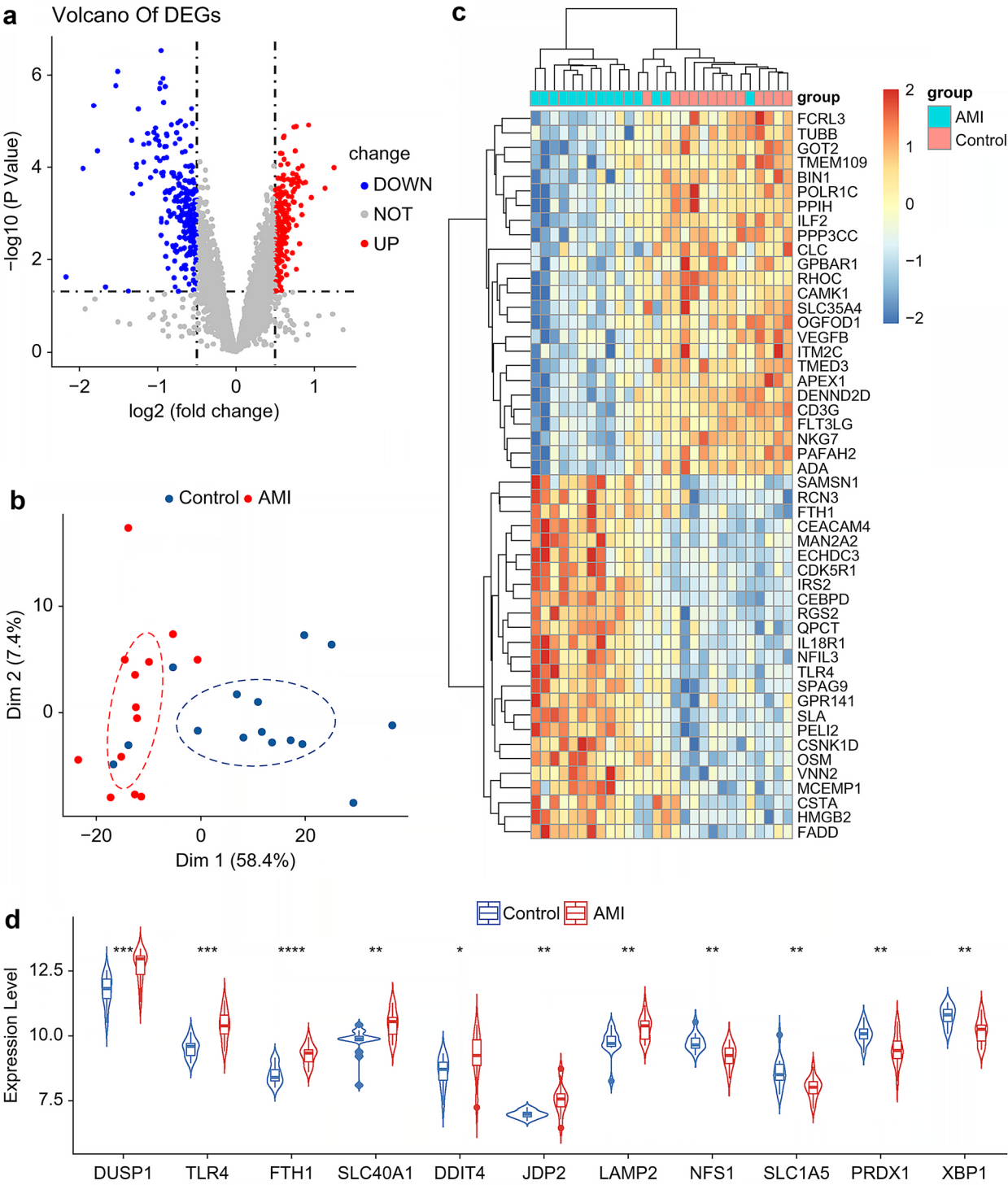


Fig. 2 Identification of key ferroptosis-related genes. **(A)** Gene volcano plots show the levels in controls and patients with AMI. **(B)** PCA for patients with AMI and controls revealed distinct expression patterns. **(C)** Heat maps showed different gene expression levels in patients with AMI and the controls. **(D)** The expression of 11 key ferroptosis-related genes differed in the controls and patients with AMI, including *DUSP1*, *TLR4*, *FTH1*, *SLC40A1*, *DDIT4*, *JDP2*, *LAMP2*, *NFS1*, *SLC1A5*, *PRDX1*, *XBP1*. AMI, acute myocardial infarction; PCA, principal component analysis. *Data source: merged data of GSE60993 and GSE29532 from GEO database

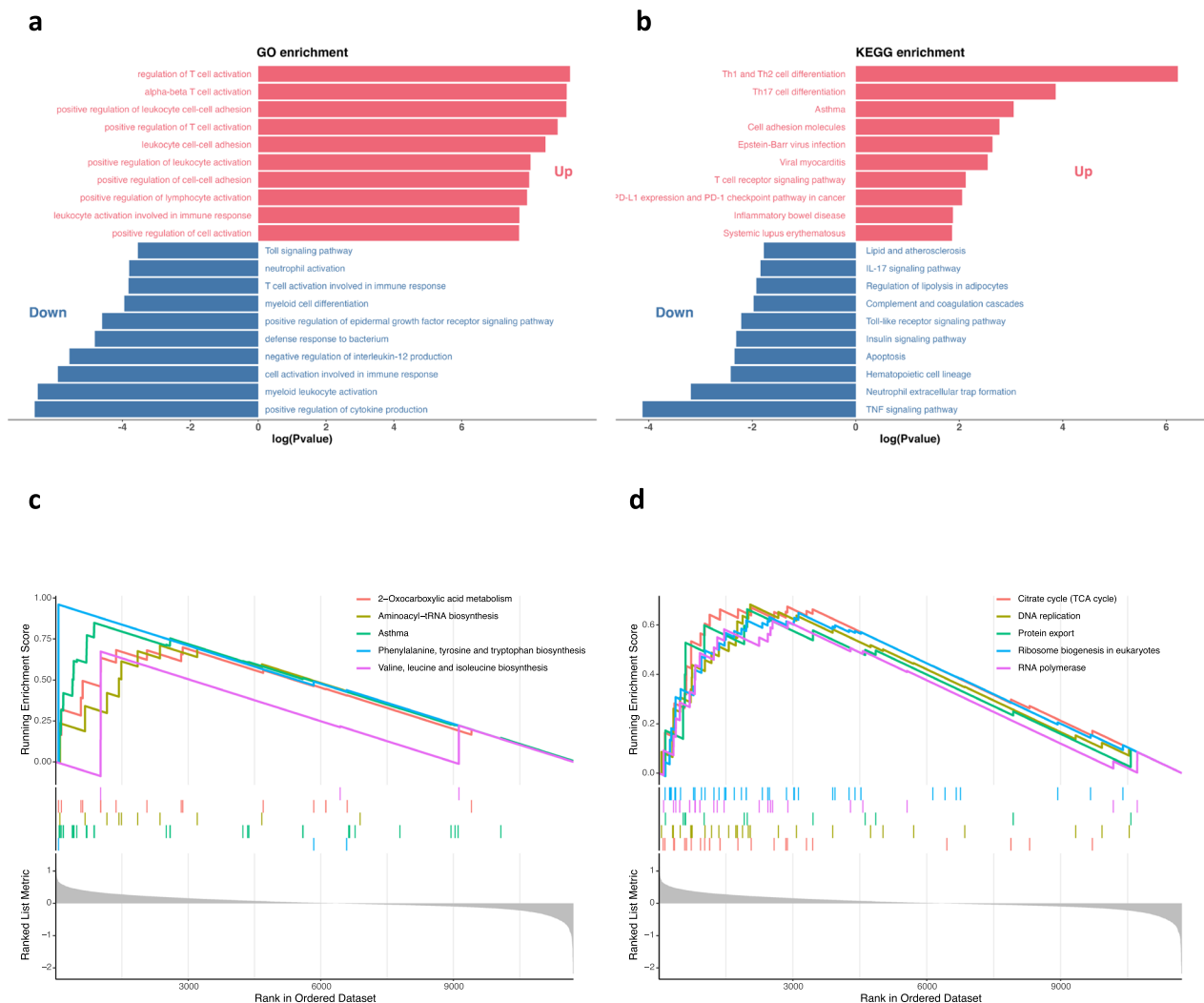
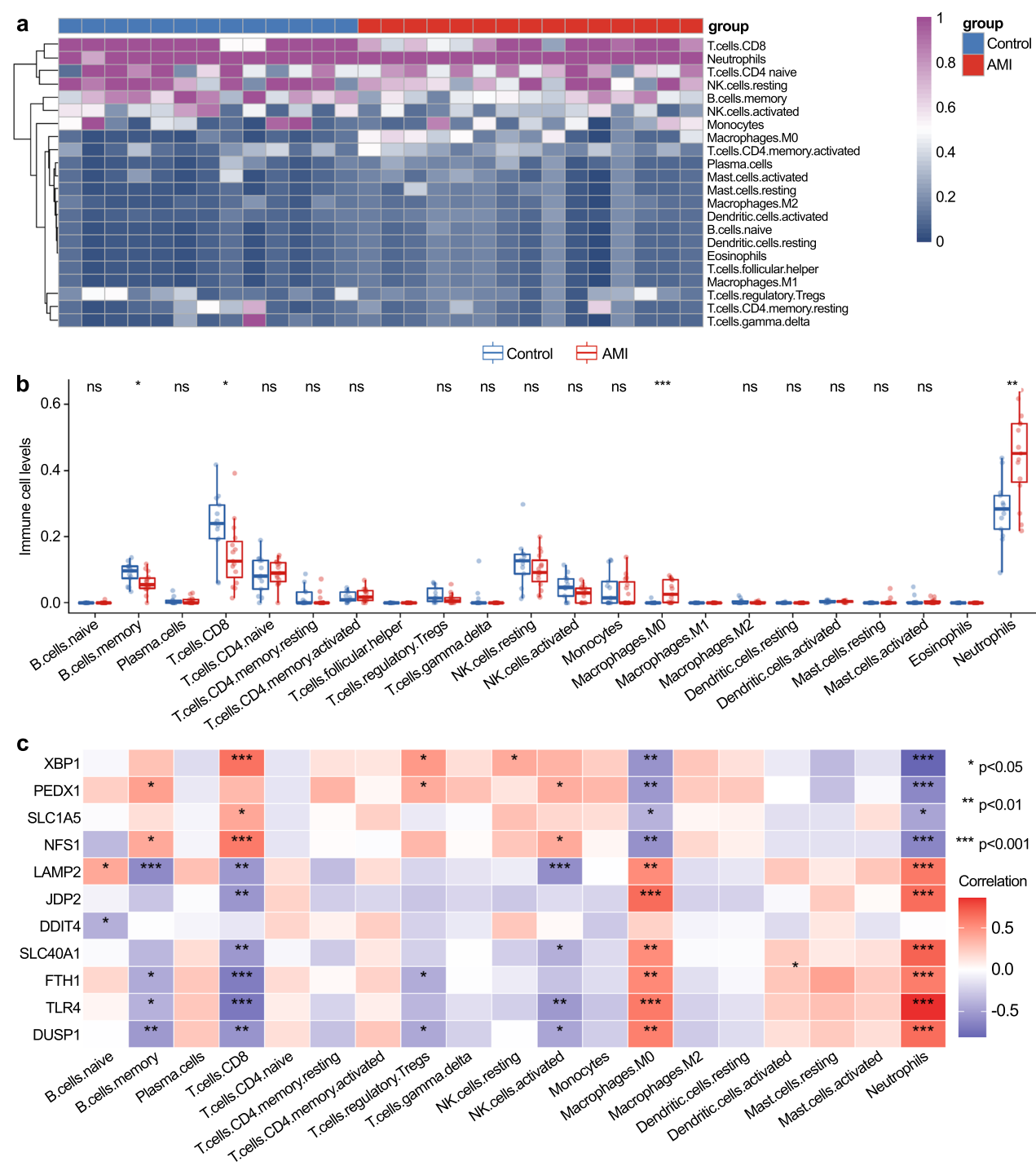


Fig. 3 Potential key signaling pathways associated with AML. **(A)** Top 10 enriched biological processes. **(B)** Top 10 pathways upregulated by KEGG. **(C-D)** Gene set enrichment analysis demonstrated the enrichment of disease related pathways. KEGG, Kyoto Encyclopedia of Genes and Genomes. *Data source: merged data of GSE60993 and GSE29532 from GEO database

findings, we conducted additional experiments to validate these upregulated genes in our clinical cohort and under cell hypoxia models, as illustrated in (Figs. 6G, H). Further analysis revealed that these 11 ferroptosis-related genes could predict long-term heart failure post-AMI. Comparison of discriminative values using PCA showed no significant differentiation with unscreened genes (Fig. 7A), whereas using the 11 key genes resulted in a clear separation (Fig. 7B), highlighting their discriminative power. In GSE59867, the predictive capability of these genes for long-term heart failure was substantial, with an AUC of 0.933 (Fig. 7C). Among these, *JD2*, *DUSP1*, and *DDIT4* demonstrated strong predictive values for long-term heart failure, with AUCs of 0.917, 0.875, and 0.736, respectively (Fig. 7D–F).

Discussion

Despite the advances in reperfusion strategies and pharmacological treatments, AMI remains a persistent global health challenge [28]. Its rapid progression often results in critical delays in treatment, leading to severe and potentially fatal outcomes. In this context, the interplay between ferroptosis and immune dysregulation in particularly complex. Ferroptosis has been demonstrated to stimulate the proliferation and activation of immune cells such as B cells, T cells, and macrophages, which play a crucial role in the immune response to AMI [29]. For example, ferroptosis initiates the migration of neutrophils to the myocardium following heart transplantation, thereby contributing to cardiac injury in mouse models [12]. Glutathione peroxidase 4 (GPX4), a key regulator in



the ferroptosis pathway, maintains cellular redox balance and significantly impacts immune responses associated with this form of cell death [30]. The release of Damage-Associated Molecular Patterns subsequent to ferroptosis can further influence the differentiation and function of immune cells in the context of tissue damage [31]. This underscores the importance of understanding and targeting ferroptosis in cardiovascular health. Gaining insights into GPX4's function within immune cells could reveal key determinants that drive ferroptosis in these cells. In this study, we identified 446 differentially expressed genes (DEGs) closely associated with immune disorders in AMI patients by analyzing the GSE60993 and GSE29532 datasets through GO and KEGG pathway analyses.

Further exploration of immune function alterations and immune cell abundance in AMI led to the identification of 11 core overlapping genes (*DUSP1*, *TLR4*, *FTH1*, *SLC40A1*, *DDIT4*, *JDP2*, *LAMP2*, *NFS1*, *SLCIA5*, *PRDX1*, and *XBP*) through a Venn diagram analysis of the 446 DEGs and 259 ferroptosis-related genes. These genes are significantly associated with immune cell infiltration, including CD8⁺ T cells, B cells, macrophages, and neutrophils, as demonstrated by correlation analysis. Additionally, these genes are linked to the inhibition of specific immune-related signaling pathways in AMI, such as interactions among lymphoid and non-lymphoid cells, primary immunodeficiency, and the adaptive immune system. Recent studies have further substantiated the connection between ferroptosis and immune cells like B cells, CD8⁺ T cells, macrophages, and neutrophils, lending further support to our findings [12, 31–33].

Furthermore, the suppression of cardiac ferroptosis by ferritin H in cardiomyocytes was recently highlighted as crucial in heart disease progression and failure [34]. Based on this, we evaluated the predictive potential of these 11 key genes for long-term heart failure in AMI patients. *JDP2*, *DUSP1*, and *DDIT4* showed excellent diagnostic ability, with AUCs of 0.917, 0.875, and 0.736, respectively. Among them, only *JDP2*, *DUSP1*, *TLR4*, *NFS1*, and *SLCIA5* were consistently identified as potential diagnostic markers across independent datasets, clinical samples, and in vitro cellular experiments.

Previous research has emphasized the essential role of macrophages in cardiac remodeling. An imbalance between pro-inflammatory M1 and anti-inflammatory

M2 macrophage phenotypes can amplify inflammation, leading to increased cardiac injury [35]. Studies have also shown a complex interaction between macrophage polarization and ferroptosis at the cellular level [31]. Our findings indicate that macrophages are positively correlated with genes such as *JDP2*, *TLR4*, and *DUSP1* but negatively correlated with *SLCIA5* and *NFS1*. We hypothesize that these identified genes could be critical mediators of the interaction between macrophage polarization and ferroptosis, influencing the pathological progression of AMI.

Among these genes, *DUSP1*, which belongs to the autophagy modulator family, is primarily involved in MAPKs within cancer cells [36]. Elevated *DUSP1* expression is associated with the suppression of autophagy-driven ferroptosis in pancreatic cancer [36]. Recent studies have also highlighted the role of cardiac *DUSP1* in reducing infarct size and improving myocardial function through the JNK pathway [37]. Additionally, the nuclear receptor coactivator 4 (NCOA4) has been found to mediate chondrocyte ferroptosis via the JNK-JUN signaling pathway [38]. However, the potential involvement of *DUSP1*-mediated ferroptosis through the JNK pathway in AMI progression remains unexplored.

JDP2, a transcription factor within the leucine zipper superfamily, regulates both transcriptional repression and activation [39]. Animal studies have shown that increased *JDP2* expression can exacerbate contractile dysfunction, induce atrial dilation and hypertrophy, and accelerate heart failure progression [40–42]. Our validation dataset supports these findings, with *JDP2* demonstrating a strong diagnostic value (AUC=0.917) for heart failure post-AMI [43]. While studies using genetically modified mice have conclusively demonstrated *JDP2*'s role in heart failure and atrial fibrillation, human studies are still lacking [39]. Further exploration of *JDP2*'s clinical implications in heart failure prediction is necessary.

DDIT4, alternatively referred to as DNA damage response 1 or stress-triggered protein, suppresses mTOR signaling by bolstering the tuberous sclerosis complex, a mechanism observed in several tumors including breast cancer, glioma, and gastric cancer [44–47]. Moreover, *DDIT4* induction by IL-10 suppresses mTOR signaling while maintaining mitochondrial integrity [48]. A recent study unveiled that *DDIT4* promotes endothelin-1 and

(See figure on next page.)

Fig. 5 Profiling of immune-related signaling pathways in patients with AMI. **(A)** Eight signaling pathways were downregulated in AMI (the negative values on the y-axis signify downregulation of the respective pathways). **(B)** The boxplot demonstrates that immunoregulatory interactions between lymphocytes and non-lymphocytes, primary immunodeficiency, and adaptive immune systems were significantly altered in the controls and the patients with AMI. **(C)** The 11 ferroptosis-related genes also showed a significant correlation with the immune-related pathways. AMI, acute myocardial infarction. *Data source: merged data of GSE60993 and GSE29532 from GEO database

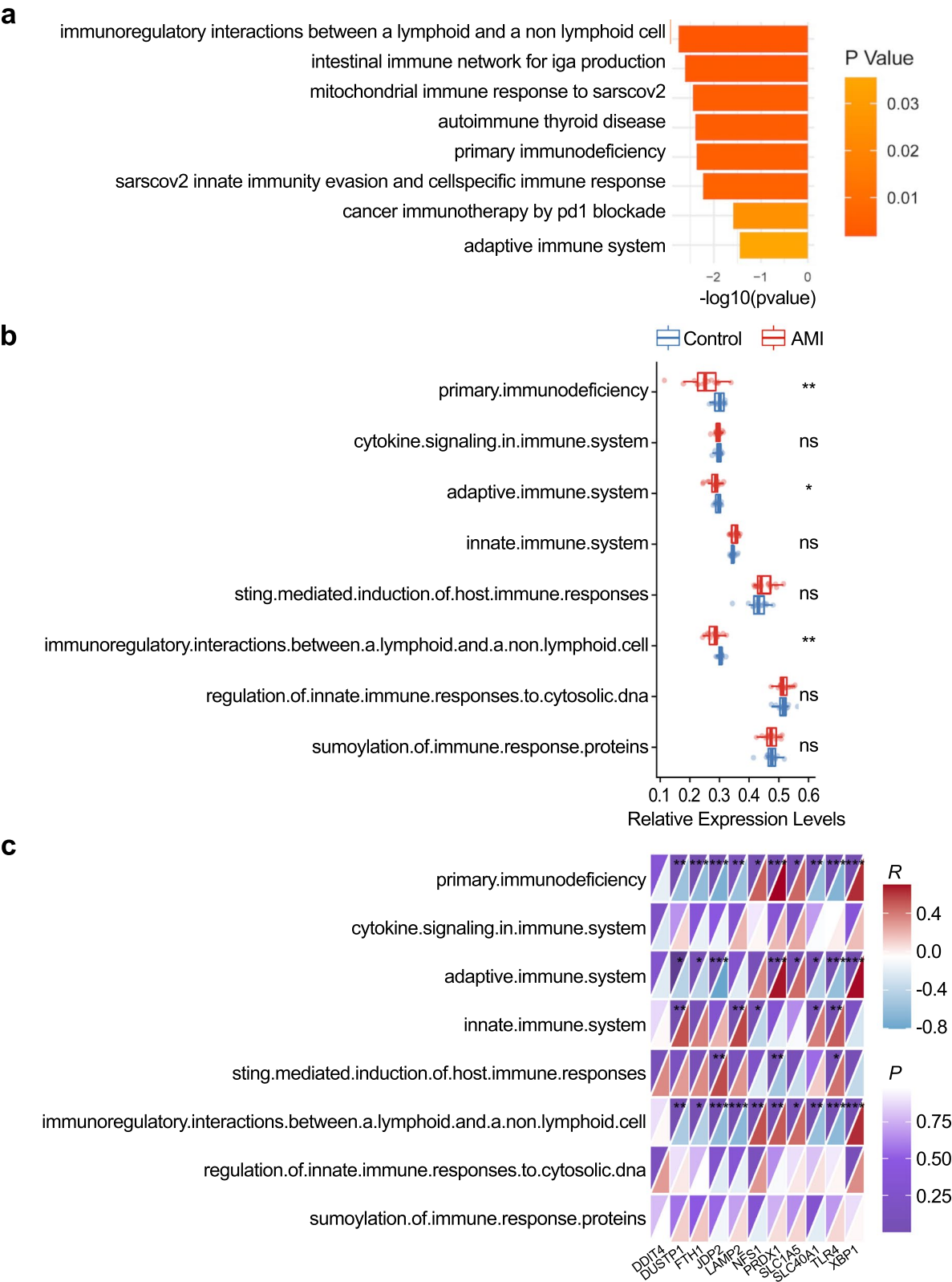


Fig. 5 (See legend on previous page.)

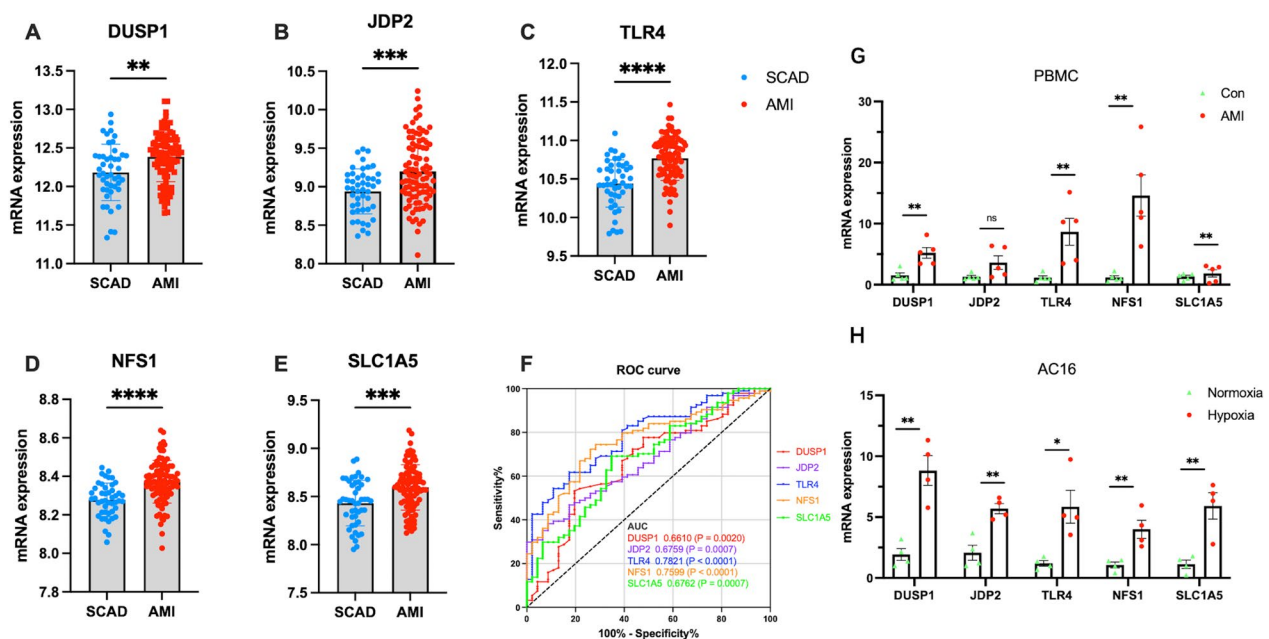


Fig. 6 Identification and assessment of the predictive values of the ferroptosis-related genes for AMI. (A–E) The relative mRNA expression levels of the *DUSP1*, *JDP2*, *TLR4*, *NFS1*, and *SLC1A5* genes in SCAD controls and patients with AMI. (F) ROC curves for *DUSP1*, *JDP2*, *TLR4*, *NFS1*, and *SLC1A5* for the SCAD controls and patients with AMI. (G) qPCR analysis of the expression patterns of five clinically correlated genes in a healthy control cohort and AMI samples from a patient cohort. (H) qPCR analysis of the expression patterns of five clinically correlated genes in AC16 cells which were treated with hypoxia 24 h. (* $p < 0.05$, ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$). *DUSP1*, dual-specificity protein phosphatase1; *JDP2*, Jun dimerization protein 2; *TLR4*, toll-like receptor 4; *DDIT4*, DNA damage-inducible transcript 4; *NFS1*, cysteine desulfurase; *SLC1A5*, neutral amino acid transporter 5; SCAD, stable coronary artery disease; AMI, acute myocardial infarction. *Data source: GSE59867 from GEO database; clinical samples; AC16 cells

hypoxia-inducible factor-1 α , contributing to injury and fibrosis in systemic lupus erythematosus end-organs [49]. However, the role of *DDIT4* in the context of hypoxia-inducible factor-1 α /mTOR pathways and their impact on long-term heart failure post-AMI remains poorly understood.

TLR4, a part of the transmembrane receptor family, is prominently found on the membranes of macrophages, dendritic cells, endothelial cells, and epithelial cells [50]. *TLR4* activation triggers pro-inflammatory responses, exacerbating inflammation and cardiac fibrosis [51]. Extracellular vesicles derived from pro-inflammatory macrophages have been shown to worsen cardiac dysfunction through the *TLR4*/NF- κ B pathway [52]. *TLR4*/Trif-centered signaling also drives inflammatory responses post-cardiac transplantation, primarily through ferroptosis and cell death mechanisms [12]. Our findings align with contemporary research linking *TLR4* to AMI [53].

NFS1 facilitates the biosynthesis of iron-sulfur clusters (Fe-S) by acting as a sulfur donor [54]. These clusters are essential cofactors for Fe-S proteins, which are involved in iron homeostasis, lipid synthesis, and energy

metabolism [55]. *NFS1* has also been shown to correlate significantly with immune cell infiltration levels [56]. However, its role in AMI has yet to be fully elucidated.

SLC1A5, often designated as ASCT2, belongs to the solute carrier family 1 and acts as a cell-surface solute-carrying transporter that regulates glutamine uptake [11]. Emerging research on the intracellular glutamine pool has highlighted their importance in protein translation, cell growth, mTORC1 signaling activation, and apoptosis prevention [57]. While *SLC1A5* amplifies ferroptosis, its function in immune cells linked to AMI remains largely unclear.

Our study highlighted six relatively unexplored ferroptosis-associated genes—*JDP2*, *DUSP1*, *DDIT4*, *TLR4*, *NFS1*, and *SLC1A5*—with potential relevance to AMI. Among these, the identified associations of *NFS1* and *SLC1A5* with AMI represent novel findings in this field. A comprehensive understanding of these genes is crucial to confirm their relevance to AMI and their potential as therapeutic targets. Despite the significant findings of this study, several limitations must be acknowledged. First, the bioinformatics approach relies on the quality of public datasets, which may introduce biases. These

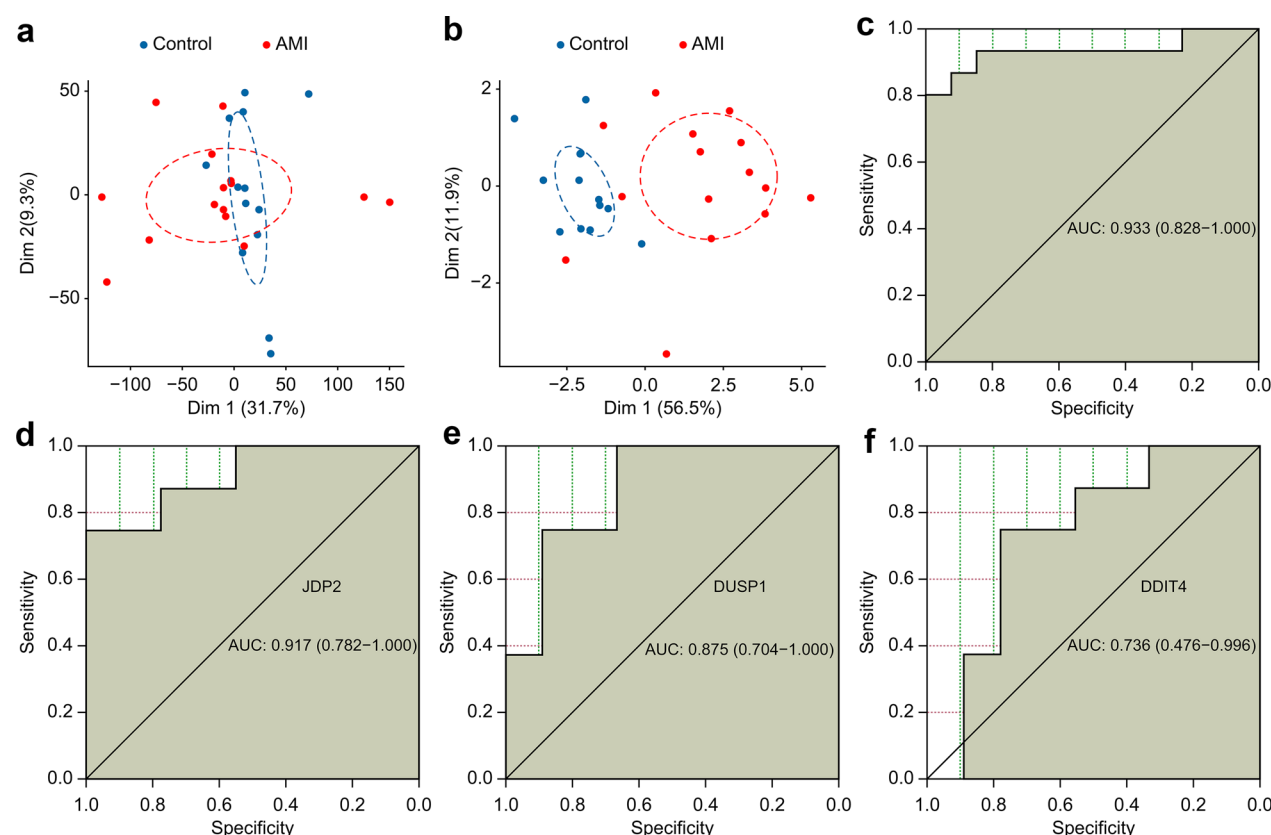


Fig. 7 Assessment of the predictive value of the ferroptosis-related genes for heart failure. **(A)** The Dim 1 distance between the controls and patients with AMI based on the unscreened genes showed no apparent difference. **(B)** A clear distinction was observed in Dim 1 based on the 11 genes. **(C)** The 11 ferroptosis genes showed a robust value for predicting long-term heart failure, with an AUC of 0.933. **(D–F)** *JDP2*, *DUSP1*, and *DDIT4* showed a robust capacity for predicting long-term heart failure, with AUCs of 0.917, 0.875, and 0.736, respectively. AUC, area under the curve; Dim 1, dimension 1; AMI, acute myocardial infarction. *DDIT4*, DNA damage-inducible transcript 4; *DUSP1*, dual-specificity protein phosphatase1; *JDP2*, Jun dimerization protein 2. *Data source: merged data of GSE60993 and GSE29532; GSE59867 from GEO database

findings must be rigorously validated through prospective, large-scale trials and functional studies in vitro and in vivo. Second, the use of the AC16 cell line as a model for ischemic conditions, although widely used, may not fully replicate the complex physiological environment of the human cardiac tissue. Additional in vivo studies or primary cardiomyocytes from human tissues would help confirm the generalizability of our findings. Third, our focus on ferroptosis-related genes addresses only one aspect of AMI pathogenesis; future research should consider other regulated cell death mechanisms, such as necroptosis and pyroptosis. These limitations should be taken into account when interpreting the results, and future work is needed to address these gaps.

Conclusions

In conclusion, the ferroptosis-related genes identified in this study are associated with AMI progression. Our research provides novel insights into the candidate

molecular interplay between immune cell infiltration and ferroptosis, and offers a foundation for exploring further investigation as potential therapeutic targets.

Abbreviations	
AMI	Acute myocardial infarction
CIBERSORT	Cell-type identification by estimating relative subsets of RNA transcript
PCA	Principal component analysis
ROC	Receiver operating characteristic
SCAD	Stable coronary artery disease
GEO	Gene expression omnibus
DEGs	Differentially expressed genes
Dim 1	First dimension
AUC	Area under the curve
GO	Gene ontology
KEGG	Kyoto Encyclopedia of genes and genomes
GSEA	Gene set enrichment analysis
PBMCs	Peripheral blood mononuclear cells
DUSP1	Dual-specificity protein phosphatase1
TLR4	Toll-like receptor 4
DDIT4	DNA damage-inducible transcript 4
JDP2	Jun dimerization protein 2
NFS1	Cysteine desulfurase
SLC1A5	Neutral amino acid transporter 5

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s40246-024-00693-7>.

Additional file1
Additional file2
Additional file3
Additional file4
Additional file5
Additional file6
Additional file7
Additional file8
Additional file9

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Author contributions

Q.Z. wrote the main manuscript text, Z.L. organized the project, R.S. contributed to the conception and revised, and Q.Z. and J.L. prepared Figs. 1–7. All authors reviewed the manuscript.

Availability of data and material

Data is provided within the manuscript or supplementary information files.

Declarations

Ethics approval and consent to participate

All experimental protocols were approved by the Ethics Committee of the Medical Ethics Committee of Xiangya Hospital of Central South University (IRB No. 2017121009). Informed consent was obtained from all participants and/or their legal guardians. All procedures were performed following the relevant guidelines and regulations, following the recommendations of the Declaration of Helsinki. All experiments were conducted according to relevant guidelines and regulations.

Competing interest

The authors declare no competing interests.

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References

1. Tsao CW, Aday AW, Almarazooq ZI, et al. Heart disease and stroke statistics-2022 update: a report from the American heart association. *Circulation*. 2022;145(8):e153–639. <https://doi.org/10.1161/CIR.0000000000001052>.
2. Roth GA, Johnson C, Abajobir A, et al. Global, regional, and national burden of cardiovascular diseases for 10 causes, 1990 to 2015. *J Am Coll Cardiol*. 2017;70(1):1–25. <https://doi.org/10.1016/j.jacc.2017.04.052>.
3. Dixon SJ, Lemberg KM, Lamprecht MR, et al. Ferroptosis: an iron-dependent form of nonapoptotic cell death. *Cell*. 2012;149(5):1060–72. <https://doi.org/10.1016/j.cell.2012.03.042>.
4. Tang D, Kroemer G. Ferroptosis. *Curr Biol*. 2020;30(21):R1292–7. <https://doi.org/10.1016/j.cub.2020.09.068>.
5. Kuang F, Liu J, Tang D, Kang R. Oxidative damage and antioxidant defense in ferroptosis. *Front Cell Dev Biol*. 2020;8: 586578. <https://doi.org/10.3389/fcell.2020.586578>.
6. Tang R, Xu J, Zhang B, et al. Ferroptosis, necroptosis, and pyroptosis in anticancer immunity. *J Hematol Oncol*. 2020;13(1):110. <https://doi.org/10.1186/s13045-020-00946-7>.
7. Xia N, Lu Y, Gu M, et al. A unique population of regulatory T cells in heart potentiates cardiac protection from myocardial infarction. *Circulation*. 2020;142(20):1956–73. <https://doi.org/10.1161/CIRCULATIONAHA.120.046789>.
8. Heidt T, Courties G, Dutta P, et al. Differential contribution of monocytes to heart macrophages in steady-state and after myocardial infarction. *Circ Res*. 2014;115(2):284–95. <https://doi.org/10.1161/CIRCRESAHA.115.303567>.
9. Peet C, Ivetic A, Bromage DI, Shah AM. Cardiac monocytes and macrophages after myocardial infarction. *Cardiovasc Res*. 2020;116(6):1101–12. <https://doi.org/10.1093/cvr/cvz336>.
10. Bulluck H, Rosmini S, Abdel-Gadir A, et al. Residual myocardial iron following intramyocardial hemorrhage during the convalescent phase of reperfused ST-segment-elevation myocardial infarction and adverse left ventricular remodeling. *Circ Cardiovasc Imaging*. 2016;9(10): e004940. <https://doi.org/10.1161/CIRCIMAGING.116.004940>.
11. Gao M, Monian P, Quadri N, Ramasamy R, Jiang X. Glutaminolysis and transferrin regulate ferroptosis. *Mol Cell*. 2015;59(2):298–308. <https://doi.org/10.1016/j.molcel.2015.06.011>.
12. Li W, Feng G, Gauthier JM, et al. Ferroptotic cell death and TLR4/Trif signaling initiate neutrophil recruitment after heart transplantation. *J Clin Invest*. 2019;129(6):2293–304. <https://doi.org/10.1172/JCI126428>.
13. Yang X, Zhu S, Li L, et al. Identification of differentially expressed genes and signaling pathways in ovarian cancer by integrated bioinformatics analysis. *Oncotargets Ther*. 2018;11:1457–74. <https://doi.org/10.2147/OTT.S152238>.
14. Fachal L, Aschard H, Beesley J, et al. Fine-mapping of 150 breast cancer risk regions identifies 191 likely target genes. *Nat Genet*. 2020;52(1):56–73. <https://doi.org/10.1038/s41588-019-0537-1>.
15. Gentles AJ, Newman AM, Liu CL, et al. The prognostic landscape of genes and infiltrating immune cells across human cancers. *Nat Med*. 2015;21(8):938–45. <https://doi.org/10.1038/nm.3909>.
16. Chen B, Khodadoust MS, Liu CL, Newman AM, Alizadeh AA. Profiling tumor infiltrating immune cells with CIBERSORT. *Methods Mol Biol*. 2018;1711:243–59. https://doi.org/10.1007/978-1-4939-7493-1_12.
17. Newman AM, Liu CL, Green MR, et al. Robust enumeration of cell subsets from tissue expression profiles. *Nat Methods*. 2015;12(5):453–7. <https://doi.org/10.1038/nmeth.3337>.
18. Al-Lamee RK, Foley M, Rajkumar CA, Francis DP. Revascularization in stable coronary artery disease. *BMJ*. 2022;377: e067085. <https://doi.org/10.1136/bmj-2021-067085>.
19. Knuuti J, Wijns W, Saraste A, et al. 2019 ESC Guidelines for the diagnosis and management of chronic coronary syndromes. *Eur Heart J*. 2020;41(3):407–77. <https://doi.org/10.1093/eurheartj/ehz425>.
20. Mandrekas JN. Receiver operating characteristic curve in diagnostic test assessment. *J Thorac Oncol*. 2010;5(9):1315–6. <https://doi.org/10.1097/JTO.0b013e3181ec173d>.
21. Yu G, Wang LG, Han Y, He QY. clusterProfiler: an R package for comparing biological themes among gene clusters. *OMICS*. 2012;16(5):284–7. <https://doi.org/10.1089/omi.2011.0118>.
22. Kanehisa M, Goto S. KEGG: kyoto encyclopedia of genes and genomes. *Nucleic Acids Res*. 2000;28(1):27–30. <https://doi.org/10.1093/nar/28.1.27>.
23. Kanehisa M. Toward understanding the origin and evolution of cellular organisms. *Protein Sci*. 2019;28(11):1947–51. <https://doi.org/10.1002/pro.3715>.
24. Kanehisa M, Furumichi M, Sato Y, Kawashima M, Ishiguro-Watanabe M. KEGG for taxonomy-based analysis of pathways and genomes. *Nucleic Acids Res*. 2023;51(D1):D587–92. <https://doi.org/10.1093/nar/gkac963>.
25. Yang M, Wang Y, He L, Shi X, Huang S. Comprehensive bioinformatics analysis reveals the role of cuproptosis-related gene Ube2d3 in myocardial infarction. *Front Immunol*. 2024;15:1353111. <https://doi.org/10.3389/fimmu.2024.1353111>.
26. Zhang L, Jia X. Down-regulation of miR-30b-5p protects cardiomyocytes against hypoxia-induced injury by targeting Aven. *Cell Mol Biol Lett*. 2019;24:61. <https://doi.org/10.1186/s11658-019-0187-4>.
27. Lee WH, Tsai MJ, Chang WA, et al. Deduction of novel genes potentially involved in hypoxic AC16 human cardiomyocytes using

- next-generation sequencing and bioinformatics approaches. *Int J Mol Med*. 2018;42(5):2489–502. <https://doi.org/10.3892/ijmm.2018.3851>.
28. Reed GW, Rossi JE, Cannon CP. Acute myocardial infarction. *Lancet*. 2017;389(10065):197–210. [https://doi.org/10.1016/S0140-6736\(16\)30677-8](https://doi.org/10.1016/S0140-6736(16)30677-8).
 29. Zhang Q, Luo Y, Peng L, et al. Ferroptosis in cardiovascular diseases: role and mechanism. *Cell Biosci*. 2023;13(1):226. <https://doi.org/10.1186/s13578-023-01169-2>.
 30. Zeng L, Yang K, Yu G, et al. Advances in research on immunocyte iron metabolism, ferroptosis, and their regulatory roles in autoimmune and autoinflammatory diseases. *Cell Death Dis*. 2024;15(7):481. <https://doi.org/10.1038/s41419-024-06807-2>.
 31. Chen X, Kang R, Kroemer G, Tang D. Ferroptosis in infection, inflammation, and immunity. *J Exp Med*. 2021. <https://doi.org/10.1084/jem.20210518>.
 32. Muri J, Thut H, Bornkamm GW, Kopf M. B1 and marginal zone B cells but not follicular B2 cells require Gpx4 to prevent lipid peroxidation and ferroptosis. *Cell Rep*. 2019;29(9):2731–44e4. <https://doi.org/10.1016/j.celrep.2019.10.070>.
 33. Liao P, Wang W, Wang W, et al. CD8(+) T cells and fatty acids orchestrate tumor ferroptosis and immunity via ACSL4. *Cancer Cell*. 2022;40(4):365–78e6. <https://doi.org/10.1016/j.ccell.2022.02.003>.
 34. Fang X, Cai Z, Wang H, et al. Loss of cardiac ferritin H facilitates cardiomyopathy via SLC7a11-mediated ferroptosis. *Circ Res*. 2020;127(4):486–501. <https://doi.org/10.1161/CIRCRESAHA.120.316509>.
 35. Mouton AJ, Li X, Hall ME, Hall JE. Obesity, hypertension, and cardiac dysfunction: novel roles of immunometabolism in macrophage activation and inflammation. *Circ Res*. 2020;126(6):789–806. <https://doi.org/10.1161/CIRCRESAHA.119.312321>.
 36. Chen X, Yu C, Kang R, Kroemer G, Tang D. Cellular degradation systems in ferroptosis. *Cell Death Differ*. 2021;28(4):1135–48. <https://doi.org/10.1038/s41418-020-00728-1>.
 37. Jin Q, Li R, Hu N, et al. DUSP1 alleviates cardiac ischemia/reperfusion injury by suppressing the Mff-required mitochondrial fission and Bnip3-related mitophagy via the JNK pathways. *Redox Biol*. 2018;14:576–87. <https://doi.org/10.1016/j.redox.2017.11.004>.
 38. Sun K, Hou L, Guo Z, et al. JNK-JUN-NCOA4 axis contributes to chondrocyte ferroptosis and aggravates osteoarthritis via ferritinophagy. *Free Radic Biol Med*. 2023;200:87–101. <https://doi.org/10.1016/j.freeradbiomed.2023.03.008>.
 39. Euler G, Kocksamper J, Schulz R, Parahuleva MS. JDP2, a novel molecular key in heart failure and atrial fibrillation? *Int J Mol Sci*. 2021. <https://doi.org/10.3390/ijms22084110>.
 40. Hill C, Wurfel A, Heger J, et al. Inhibition of AP-1 signaling by JDP2 overexpression protects cardiomyocytes against hypertrophy and apoptosis induction. *Cardiovasc Res*. 2013;99(1):121–8. <https://doi.org/10.1093/cvr/cvt094>.
 41. Heger J, Bornbaum J, Wurfel A, et al. JDP2 overexpression provokes cardiac dysfunction in mice. *Sci Rep*. 2018;8(1):7647. <https://doi.org/10.1038/s41598-018-26052-w>.
 42. Parahuleva MS, Kocksamper J, Heger J, et al. Structural, pro-inflammatory and calcium handling remodeling underlies spontaneous onset of paroxysmal atrial fibrillation in JDP2-overexpressing mice. *Int J Mol Sci*. 2020. <https://doi.org/10.3390/ijms21239095>.
 43. Maciejak A, Kiliszek M, Michalak M, et al. Gene expression profiling reveals potential prognostic biomarkers associated with the progression of heart failure. *Genome Med*. 2015;7(1):26. <https://doi.org/10.1186/s13073-015-0149-z>.
 44. Zhang Y, Liu X, Wang Y, et al. The m(6)A demethylase ALKBH5-mediated upregulation of DDIT4-AS1 maintains pancreatic cancer stemness and suppresses chemosensitivity by activating the mTOR pathway. *Mol Cancer*. 2022;21(1):174. <https://doi.org/10.1186/s12943-022-01647-0>.
 45. Miao ZF, Sun JX, Adkins-Threats M, et al. DDIT4 licenses only healthy cells to proliferate during injury-induced metaplasia. *Gastroenterology*. 2021;160(1):260–71e10. <https://doi.org/10.1053/j.gastro.2020.09.016>.
 46. Horak P, Crawford AR, Vadysirisack DD, et al. Negative feedback control of HIF-1 through REDD1-regulated ROS suppresses tumorigenesis. *Proc Natl Acad Sci USA*. 2010;107(10):4675–80. <https://doi.org/10.1073/pnas.0907705107>.
 47. Ho KH, Chen PH, Chou CM, et al. A key role of DNA damage-inducible transcript 4 (DDIT4) connects autophagy and GLUT3-mediated stemness to desensitize temozolomide efficacy in glioblastomas. *Neurotherapeutics*. 2020;17(3):1212–27. <https://doi.org/10.1007/s13311-019-00826-0>.
 48. Ip WKE, Hoshi N, Shouval DS, Snapper S, Medzhitov R. Anti-inflammatory effect of IL-10 mediated by metabolic reprogramming of macrophages. *Science*. 2017;356(6337):513–9. <https://doi.org/10.1126/science.aal3535>.
 49. Frangou E, Chrysanthopoulou A, Mitsios A, et al. REDD1/autophagy pathway promotes thromboinflammation and fibrosis in human systemic lupus erythematosus (SLE) through NETs decorated with tissue factor (TF) and interleukin-17A (IL-17A). *Ann Rheum Dis*. 2019;78(2):238–48. <https://doi.org/10.1136/annrheumdis-2018-213181>.
 50. Zhang Q, Wang L, Wang S, et al. Signaling pathways and targeted therapy for myocardial infarction. *Signal Transduct Target Ther*. 2022;7(1):78. <https://doi.org/10.1038/s41392-022-00925-z>.
 51. Xu GR, Zhang C, Yang HX, et al. Modified citrus pectin ameliorates myocardial fibrosis and inflammation via suppressing galectin-3 and TLR4/MyD88/NF-kappaB signaling pathway. *Biomed Pharmacother*. 2020;126:110071. <https://doi.org/10.1016/j.biopha.2020.110071>.
 52. Biemmi V, Milano G, Ciullo A, et al. Inflammatory extracellular vesicles prompt heart dysfunction via TLR4-dependent NF-kappaB activation. *Theranostics*. 2020;10(6):2773–90. <https://doi.org/10.7150/thno.39072>.
 53. Wu J, Cai H, Lei Z, et al. Expression pattern and diagnostic value of ferroptosis-related genes in acute myocardial infarction. *Front Cardiovasc Med*. 2022;9:993592. <https://doi.org/10.3389/fcvm.2022.993592>.
 54. Lin JF, Hu PS, Wang YY, et al. Phosphorylated NF51 weakens oxaliplatin-based chemosensitivity of colorectal cancer by preventing PANoptosis. *Signal Transduct Target Ther*. 2022;7(1):54. <https://doi.org/10.1038/s41392-022-00889-0>.
 55. Alvarez SW, Sviderskiy VO, Terzi EM, et al. NF51 undergoes positive selection in lung tumours and protects cells from ferroptosis. *Nature*. 2017;551(7682):639–43. <https://doi.org/10.1038/nature24637>.
 56. Lv Z, Wang J, Wang X, et al. Identifying a ferroptosis-related gene signature for predicting biochemical recurrence of prostate cancer. *Front Cell Dev Biol*. 2021;9:666025. <https://doi.org/10.3389/fcell.2021.666025>.
 57. van Geldermalsen M, Wang Q, Nagarajah R, et al. ASCT2/SLC1A5 controls glutamine uptake and tumour growth in triple-negative basal-like breast cancer. *Oncogene*. 2016;35(24):3201–8. <https://doi.org/10.1038/onc.2015.381>.

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